

Topic 1: Basic Linux Commands

1. **Pwd** ~ Display the current working directory

```
ketana_sindhu@KetaZone:~$ pwd
/home/ketana_sindhu
ketana_sindhu@KetaZone:~$ |
```

2. **Clear** ~ clear the terminal screen

```
ketana_sindhu@KetaZone:~$ |
```

3. **History** ~ displays previously executed commands

```
150  pwd
151  clear
152  history
ketana_sindhu@KetaZone:~$ |
```

4. **Whoami** ~ display the current logged-in username

```
ketana_sindhu@KetaZone:~$ whoami
ketana_sindhu
ketana_sindhu@KetaZone:~$ |
```

5. **Date** ~ display the current system date and time

```
ketana_sindhu@KetaZone:~$ date
Tue Aug  4 04:13:02 UTC 2026
ketana_sindhu@KetaZone:~$ |
```

6. **Hostname** ~ Display the system hostname

```
ketana_sindhu@KetaZone:~$ hostname
KetaZone
ketana_sindhu@KetaZone:~$ |
```

7. **Uname -a** ~ Display operating system operation

```
ketana_sindhu@KetaZone:~$ uname -a
Linux KetaZone 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
ketana_sindhu@KetaZone:~$ |
```

8. **Exit** ~ exit the terminal

```
ketana_sindhu@KetaZone:~$ exit|
```

TOPIC 2 ~ DIRECTORY MANAGEMENT

1. Ls ~ list files and directories

```
ketana_sindhu@KetaZone:~$ ls
DisplayProcessID.c  File.c      FirstFile  Hello.c    ParentAndChild.c  S2Commands  WAIT      example.txt  hello.c  ketana_sindhu  pid  wait
File               FirstFile  Fork.c     PID.c      ReadUser.c        Sindhu      Wait.c    file         ketana   name          sindhu  wait.c
ketana_sindhu@KetaZone:~$ |
```

2. Ls -a ~ display detailed file information

```
ketana_sindhu@KetaZone:~$ ls -a
.          .bash_logout  .config    .motd_shown  File      FirstFile  PID.c      ParentAndChild.c  S2Commands  Wait.c      hello.c      name  wait
..         .bashrc      .landscape .profile     File.c    Fork.c     ParentAndChild.c  Sindhu      example.txt  ketana       pid    wait.c
.bash_history .cache       .local     DisplayProcessID.c FirstFile  Hello.c     ReadUser.c        WAIT      file         ketana_sindhu  sindhu
ketana_sindhu@KetaZone:~$ |
```

3. Cd ~ change the current directory

```
ketana_sindhu@KetaZone:~$ cd
ketana_sindhu@KetaZone:~$ |
```

4. Cd .. ~ move to parent directory

```
ketana_sindhu@KetaZone:~$ cd ..
ketana_sindhu@KetaZone:/home$ |
```

5. mkdir ~ create a new directory

```
ketana_sindhu@KetaZone:/home$ mkdir OSLAB
```

6. rmdir ~ remove a directory

```
ketana_sindhu@KetaZone:/home$ rmdir OSLAB
```

7. find FileName.c ~ search for files and directories

```
ketana_sindhu@KetaZone:/home$ find Hello.c
```

8. locate ~ Quickly locate files

```
ketana_sindhu@KetaZone:~$ locate Hello.c
```

3.FILE MANAGEMENT

1.touch ~ Create new empty file

```
ketana_sindhu@KetaZone:~$ touch KL.c
```

2. cat ~ display file contents

```
ketana_sindhu@KetaZone:~$ cat Hello.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
ketana_sindhu@KetaZone:~$ |
```

3. cp ~ copy files or directories

```
ketana_sindhu@KetaZone:~$ cp Hello.c FirstFile.c
ketana_sindhu@KetaZone:~$ |
```

4. mv ~ move or rename files

```
ketana_sindhu@KetaZone:~$ mv Hello.c hello.c
ketana_sindhu@KetaZone:~$ |
```

5. rm ~ remove file

```
ketana_sindhu@KetaZone:~$ rm hello.c
ketana_sindhu@KetaZone:~$ |
```

6. head ~ display first lines of a file

```
ketana_sindhu@KetaZone:~$ head FirstFile.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
ketana_sindhu@KetaZone:~$ |
```

7. tail ~ display last lines of the file

```
ketana_sindhu@KetaZone:~$ tail FirstFile.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
ketana_sindhu@KetaZone:~$ |
```

8.more ~ view a file page by page

```
ketana_sindhu@KetaZone:~$ more FirstFile.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
ketana_sindhu@KetaZone:~$ |
```

9.less ~ Scroll through a file

```
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
FirstFile.c (END)
```

10.wc ~ count lines, words and characters

```
ketana_sindhu@KetaZone:~$ wc FirstFile.c
 7  9 77 FirstFile.c
ketana_sindhu@KetaZone:~$ |
```

11.diff ~ compare two files

```
ketana_sindhu@KetaZone:~$ diff FirstFile.c PID.c
1a2
> #include<unistd.h>
5c6,19
<     printf("Hello Linux\n");
---
>     pid_t pid;
>
>     pid=fork();
>
>     if(pid==0)
>     {
>         printf("Child PID = %d\n",getpid());
>         printf("Parent PID = %d\n",getppid());
>     }
>     else
>     {
>         printf("Parent PID = %d\n",getpid());
>     }
>
ketana_sindhu@KetaZone:~$ |
```

12.sort ~ sort file contents

```
ketana_sindhu@KetaZone:~$ sort FirstFile.c

    printf("Hello Linux\n");
    return 0;
#include<stdio.h>
int main()
{
}
ketana_sindhu@KetaZone:~$ |
```

13.uniq ~ remove duplicate files

```
ketana_sindhu@KetaZone:~$ uniq FirstFile.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
ketana_sindhu@KetaZone:~$ |
```

14.grep ~ search for text inside a file

```
ketana_sindhu@KetaZone:~$ grep main FirstFile.c
int main()
ketana_sindhu@KetaZone:~$ |
```

4. TOPIC ~ FILE EDITING

1.nano ~ create or edit a file

```
ketana_sindhu@KetaZone:~$ nano Hello.c
ketana_sindhu@KetaZone:~$ |
```

2. vi ~ edit a file using vim editor

```
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

5. C PROGRAM COMPILATION AND EXECUTION

1. gcc ~ compile a C program

```
ketana_sindhu@KetaZone:~$ gcc FirstFile.c -o file
```

2. ./program ~ Execute the compiled program

```
ketana_sindhu@KetaZone:~$ gcc FirstFile.c -o file
ketana_sindhu@KetaZone:~$ ./file
Hello Linux
```

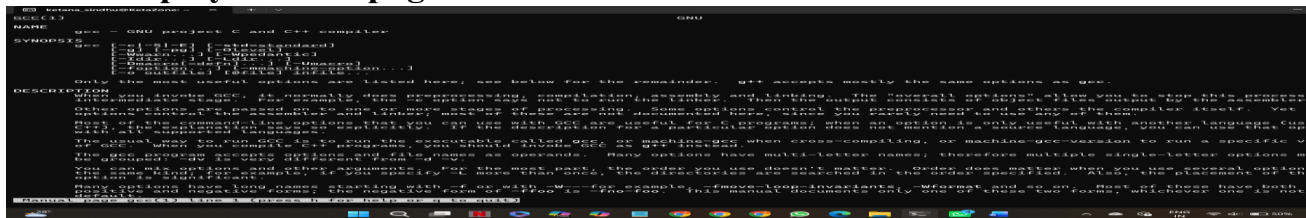
3. which ~ Locate the executable of a command

```
ketana_sindhu@KetaZone:~$ which gcc
/usr/bin/gcc
ketana_sindhu@KetaZone:~$ |
```

4. whereis ~ Locate binary, source and manual pages

```
ketana_sindhu@KetaZone:~$ whereis gcc
gcc: /usr/bin/gcc /usr/lib/gcc /usr/libexec/gcc /usr/share/gcc /usr/share/man/man1/gcc
ketana_sindhu@KetaZone:~$ |
```

5. man ~ display manual pages



6.PROCESS MANAGEMENT

1. ps ~ display running processes

```
ketana_sindhu@KetaZone:~$ ps
  PID TTY          TIME CMD
  343 pts/0        00:00:00 bash
  467 pts/0        00:00:00 ps
ketana_sindhu@KetaZone:~$
```

2. `ps -ef ~` display all running processes

```
ketana_sindhu@KetaZone:~$ ps -ef
```

UID	PID	PPID	C	STIME	TTY	TIME	CMD
root	1	0	1	15:31	?	00:00:00	/sbin/init
root	2	1	0	15:31	hvc0	00:00:00	/init
root	6	2	0	15:31	hvc0	00:00:00	plan9 -
root	46	1	0	15:31	?	00:00:00	/usr/li
systemd+	82	1	0	15:31	?	00:00:00	/usr/li
root	89	1	0	15:31	?	00:00:00	/usr/li
root	144	1	0	15:31	?	00:00:00	/bin/sh
root	151	1	0	15:31	?	00:00:00	/usr/st
message+	160	1	0	15:31	?	00:00:00	@dbus-d
root	165	1	0	15:31	?	00:00:00	/usr/bi

3. top ~ monitor running processes

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMA
1	root	0	0	2445556	1500000	11061	0	0.0	0.0	0.00	0.00
4	root	0	0	3180000	1500000	194048	0	0.0	0.0	0.00	0.00
46	root	0	0	4322200	165676	154416	0	0.0	0.0	0.00	0.00
138	root	0	0	3544300	1500000	90037	0	0.0	0.0	0.00	0.00
144	root	0	0	3888000	19444	1832	0	0.0	0.0	0.00	0.00
151	root	0	0	44477	3800000	28444	0	0.0	0.0	0.00	0.00
156	root	0	0	3888000	1500000	90037	0	0.0	0.0	0.00	0.00
165	root	0	0	4440000	207600	146160	0	0.0	0.0	0.00	0.00
174	root	0	0	184332	94008	8168	0	0.0	0.0	0.00	0.00
177	root	0	0	33666	1500000	125448	0	0.0	0.0	0.00	0.00
187	root	0	0	200442	44444	55664	0	0.0	0.0	0.00	0.00
195	root	0	0	200442	33666	55664	0	0.0	0.0	0.00	0.00
200	root	0	0	33666	33666	180004	0	0.0	0.0	0.00	0.00
207	root	0	0	33666	33666	117884	0	0.0	0.0	0.00	0.00
216	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
220	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
223	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
224	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
225	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
226	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
227	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
228	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
229	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
230	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
231	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
232	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
233	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
234	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
235	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
236	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
237	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
238	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
239	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
240	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
241	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
242	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
243	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
244	root	0	0	33666	33666	111112	0	0.0	0.0	0.00	0.00
245	root	0	0	33666	33666	111112	0	0.0			

4. kill ~ terminate a process

```
ketana_sindhu@KetaZone:~$ kill 1234
-bash: kill: (1234) - No such process
```

5. kill -9 ~ Forcefully terminate a process

```
ketana_sindhu@KetaZone:~$ kill -9  
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]  
ketana_sindhu@KetaZone:~$ |
```

7. JOB CONTROL

1. jobs ~ Display background jobs

```
ketana_sindhu@KetaZone:~$ jobs  
ketana_sindhu@KetaZone:~$ |
```

2. bg ~ Resume a job in the background

```
ketana_sindhu@KetaZone:~$ bg %1  
-bash: bg: %1: no such job  
ketana_sindhu@KetaZone:~$ |
```

3. fg ~ Bring a job to the foreground

```
ketana_sindhu@KetaZone:~$ fg %1  
-bash: fg: %1: no such job  
ketana_sindhu@KetaZone:~$ |
```

4. & - Run a process in the background

```
ketana_sindhu@KetaZone:~$ ./FirstFile &  
[1] 801  
ketana_sindhu@KetaZone:~$ Hello Linux  
|
```


8. FILE PERMISSIONS AND LINKS

1. chmod ~ Change file permissions

```
ketana_sindhu@KetaZone:~$ chmod +x script.sh
```

2. chmod 755 ~ Set read, write and execute permissions

```
ketana_sindhu@KetaZone:~$ chmod 755
```

3. ln ~ create a hard link

```
ketana_sindhu@KetaZone:~$ ln FirstFile.c HELLO.c
ketana_sindhu@KetaZone:~$ |
```

4. ln -s ~ create a symbolic link

```
ketana_sindhu@KetaZone:~$ ln -s FirstFile shortcut
ketana_sindhu@KetaZone:~$ |
```

9. DISK AND MEMORY INFORMATION

1. df -h ~ display disk space usage

```
ketana_sindhu@KetaZone:~$ df -h
Filesystem      Size      Used Avail Use% Mounted on
none            3.8G      0      3.8G   0% /usr/lib/modules/
none            3.8G      0      3.8G   0% /mnt/wsl/
drivers         476G     141G     335G   30% /usr/lib/wsl/driv
/dev/sdd        1007G     1.7G     995G    1% /
none            3.8G      0      3.8G   0% /mnt/wslg
none            3.8G      0      3.8G   0% /usr/lib/wsl/lib
rootfs          3.8G      0      3.8G   0% /run
rootfs          3.8G      0      3.8G   0% /run/lock
rootfs          3.8G      0      3.8G   0% /run/shm
rootfs          3.8G      0      3.8G   0% /mnt/wslg/versions
rootfs          3.8G      0      3.8G   0% /mnt/wslg/doc
C:\             476G     141G     335G   30% /mnt/c
tmpfs           3.8G      0      3.8G   0% /tmp
none            1.0M      0      1.0M   0% /run/credentials/
none            1.0M      0      1.0M   0% /run/credentials/
none            1.0M      0      1.0M   0% /run/credentials/
tmpfs           766M     12K     766M    1% /run/user/1000
ketana_sindhu@KetaZone:~$ |
```

2. du -sh ~ display directory size

```
ketana_sindhu@KetaZone:~$ du -sh
352K .
ketana_sindhu@KetaZone:~$ |
```

3. free -h ~ display memory usage (Linux)

```
ketana_sindhu@KetaZone:~$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	7.5Gi	594Mi	6.8Gi	3.5Mi	217Mi	6.9Gi
Swap:	2.0Gi	0B	2.0Gi			

```
ketana_sindhu@KetaZone:~$
```

10.USER AND ENVIRONMENT MANAGEMENT

1.id ~ display user and group IDs

```
ketana_sindhu@KetaZone:~$ id
uid=1000(ketana_sindhu) gid=1000(ketana_sindhu) groups=1000(ketana_sindhu),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(us
```

2.env ~ display environment variables

[illegible]

3.export ~ Set an environment variable

```
ketana_sindhu@KetaZone:~$ export PATH=$PATH:..
ketana_sindhu@KetaZone:~$
```

4. alias ~ Create a command shortcut

```
ketana_sindhu@KetaZone:~$ alias ll='ls -l'
ketana_sindhu@KetaZone:~$
```